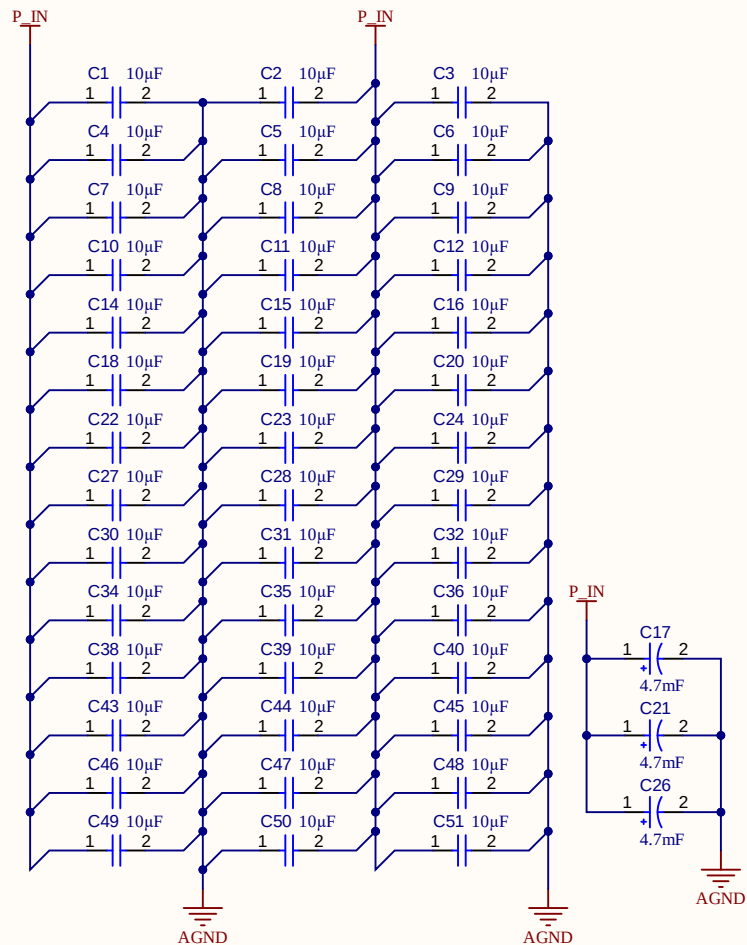


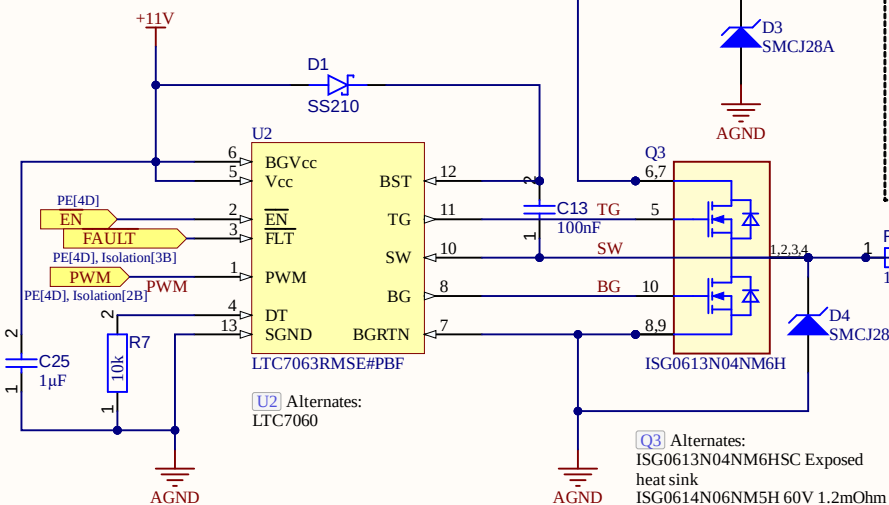
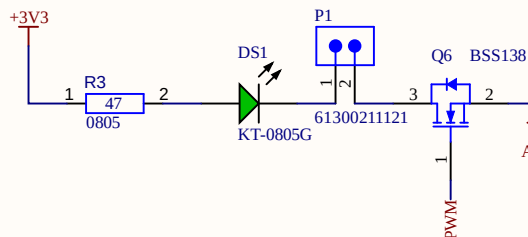
Bulk Capacitance

40 x 10 μ F 1210 X7S capacitors
200 μ F Capacitance at 30V DC bias



[C17] [C21] [C26]

Also contain footprints
for 2x5mm electrolytic
capacitors

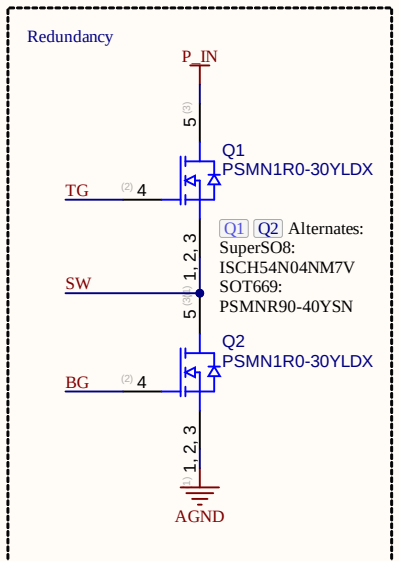


Jumper to REF
 $V(L_{sense}) = I \cdot 0.05 + 0.6125$

$I_{out} = (V_{sense} - 0.6125) / 0.05$

Jumper to AGND
 $V(L_{sense}) = I \cdot 0.05$

$I_{out} = (V_{sense}) / 0.05$



[Q1] [Q2] Alternates:
SuperSO8:
ISCH54N04NM7V
SOT669:
PSMNR90-40YSN

[U3] Alternate:
INA241A3IDR, INA241B3ID
R: Av:50
INA241A4IDR: Av:100
TSC2011IDT: Av:60 - Pin 4
shutdown

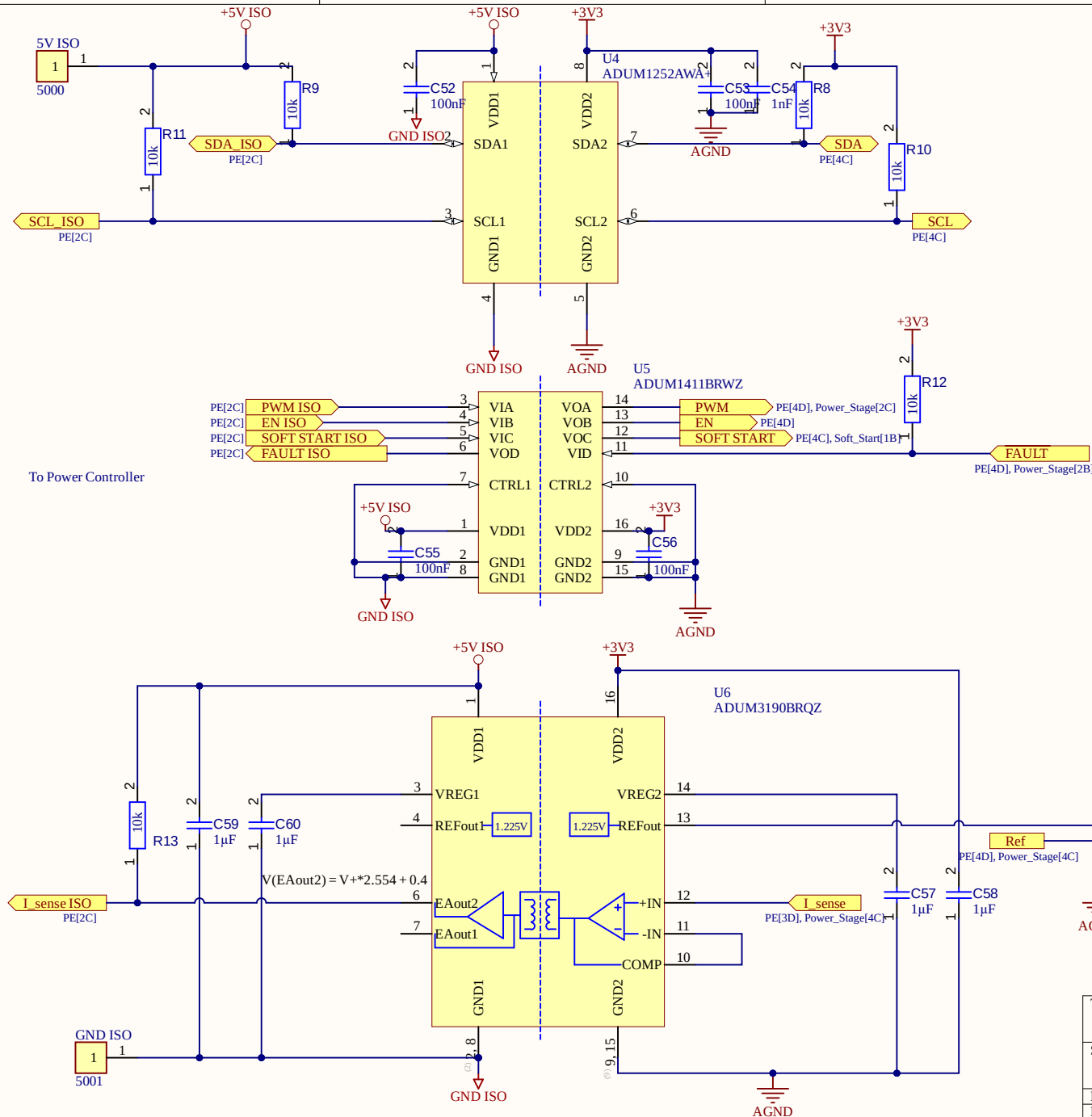
Title		
Half Bridge Schematic For Geec Power Electronics		
Size	Number	Revision
A4		
Date:	3/22/2026	Sheet 2 of 6
File:	C:\Users\...\Power_Stage.SchDoc	Drawn By: Conor Farrell

Alternate Parts:

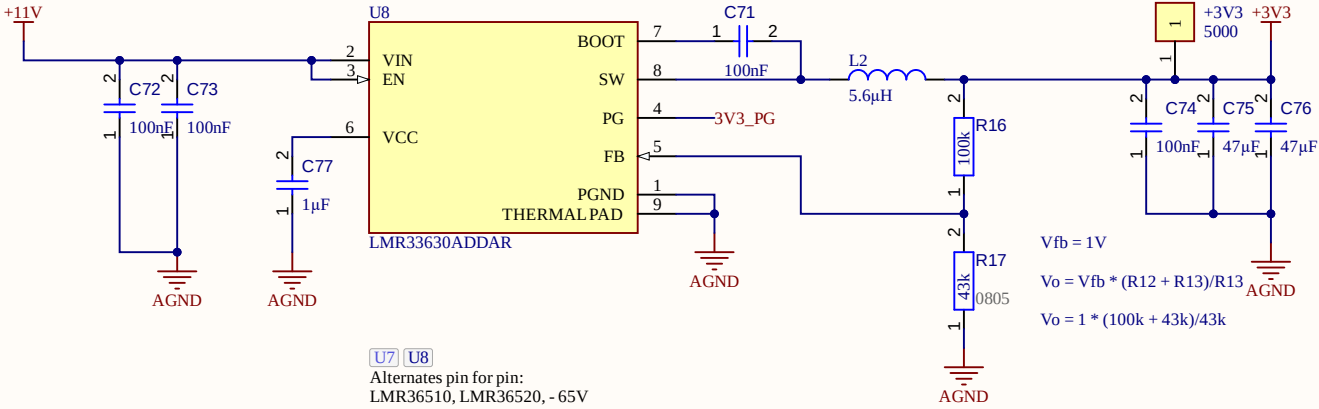
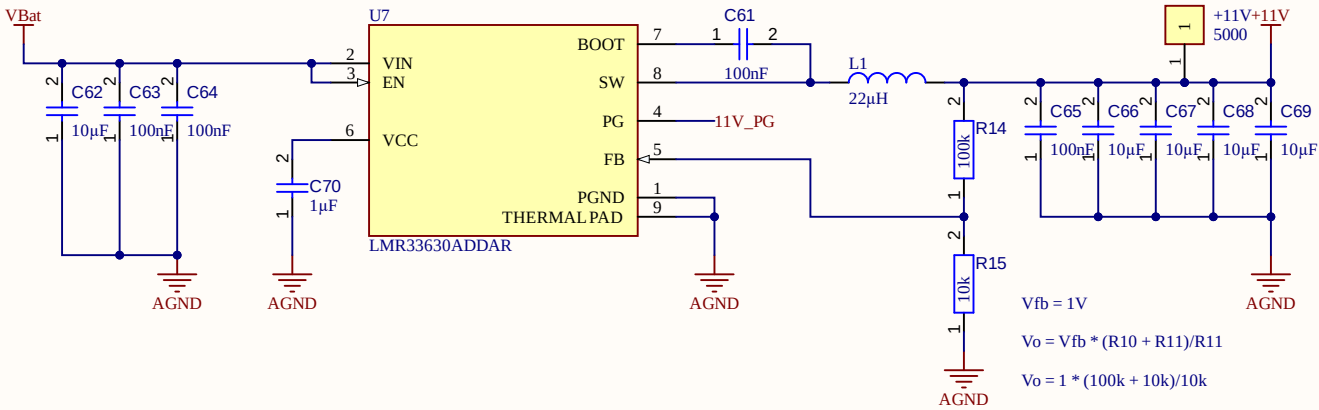
U4 ADuM1252: ADuM1254, ISO1540, ISO1640

U5 ADuM1411: ADuM1410 will work but
FAULT Won't work
ISO6741 requires PIN 10 and 7 disconnected
Other SOIC TI 3 pin forward 1 reverse will similarly work

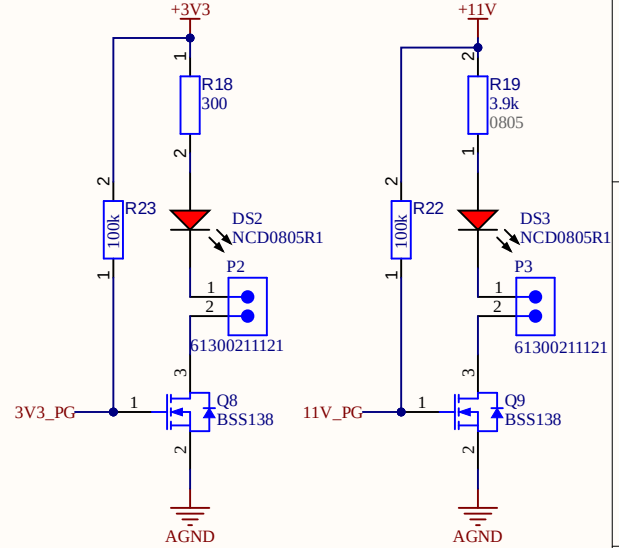
U6 ADuM3190B - ADuM3190T 400kHz version,
ADuM3190A is 200kHz and could work



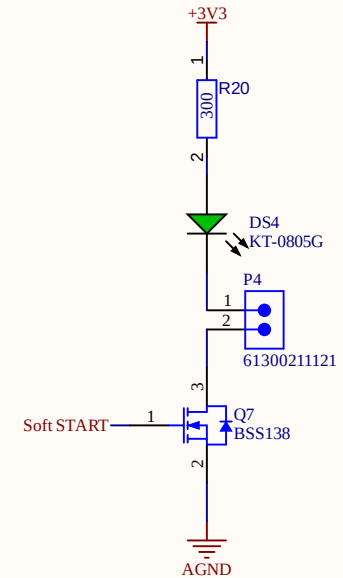
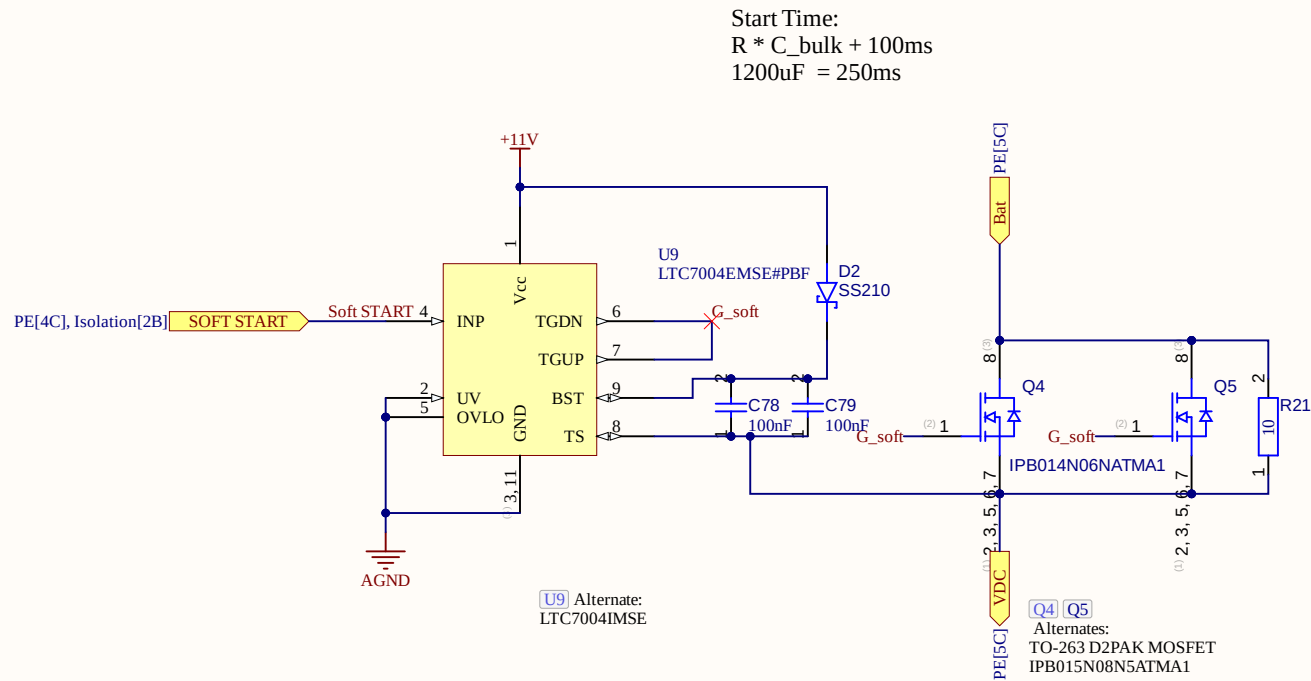
Title		
Isolation For Geec Power Electronics		
Size	Number	Revision
A4		
Date:	3/22/2026	Sheet 3 of 6
File:	C:\Users\...\Isolation.SchDoc	Drawn By: Conor Farrell



U7 U8
Alternates pin for pin:
LMR36510, LMR36520, - 65V
rated 1A and 2A
LMR33610, LMR33620,
LMR33640 - 36V rated 1A, 2A,
4A



Title		
Half Bridge Schematic For Geece Power Electronics		
Size	Number	Revision
A4		
Date:	3/22/2026	Sheet 4 of 6
File:	C:\Users\...\Power_Supply.SchDoc	Drawn By: Conor Farrell



Title Soft Start For Geec Power Electronics		
Size A4	Number	Revision
Date: 3/22/2026	Sheet 5 of 6	
File: C:\Users\...\Soft_Start.SchDoc	Drawn By: *	

